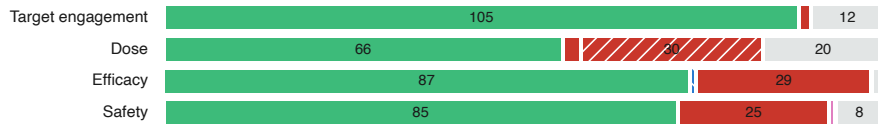
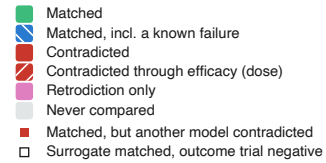
a





Panel b: every tile; diseases by global DALYs in 2021 (WHO Global Health Estimates), then those without a WHO cause by area.

b

		Target engagement	Dose	Efficacy	Safety
#1	**type 2 diabetes mellitus** GLP1R agonists (12)	**2011** Cynomolgus monkeys and rat islets (dulaglutide)	**2019** Beagle dog oral bioavailability of semaglutide/SNAC tablets	**2022** GLP1R Gly168Ser genotype, GLP-1 infusion clamp in healthy volunteers	**2016** GLP1R Ala316Thr carriers (human genetics)
	type 2 diabetes mellitus GCK activators (9)	**2011** Rodent models of type 2 diabetes	**2012** Glucose-insulin-glucagon model (LY2599506) Dose reached; effect faded	**2019** People with GCK-activating or GKRP loss-of-function variants AZD1656 missed 4-month HbA1c	**2023** Rodent models of type 2 diabetes GCK proxy misses GKA lipid rise
	type 2 diabetes mellitus HSD11B1 inhibitors (8)	**2021** DIO mice, 11β-HSD1 inhibitor Adipose inhibition held 14 days	**2015** DIO mice, 11β-HSD1 inhibitor Full inhibition reached; no glucose effect	**2015** DIO mice, 11β-HSD1 inhibitor Complete inhibition, no glucose lowering	**2022** Cortisone reductase deficiency: people with loss-of-function HSD11B1 or H6PD
	type 2 diabetes mellitus DPP4 inhibitors (7)	**2016** Rodent plasma DPP-4 inhibition and oral glucose load (Wistar, Zucker fatty, DIO, db/db)	**2016** Rodent plasma DPP-4 inhibition and oral glucose load (Wistar, Zucker fatty, DIO, db/db)	**2023** Population PK/PD plus meta-analysis, ofrogliptin single dose to DPP-4 inhibition to HbA1c	**2016** Concentration-QTc model, omarigliptin single-dose study Monkey skin harm not in patients
	type 2 diabetes mellitus INSR agonists (6)	**2023** STZ-diabetic rat single-dose PK/glucodynamics (Lilly)	**2022** STZ-diabetic rat single-dose PK/glucodynamics (Lilly)	**2022** STZ-diabetic rat single-dose PK/glucodynamics (Lilly)	**2015** In vitro mitogenicity in L6-hIR, HMEC and Saos-2 cells Rat/dog tox missed liver fat, ALT
	type 2 diabetes mellitus GCGR/GLP1R agonists (5)	**2023** Human GLP-1 + glucagon co-infusion challenge (overweight volunteers without diabetes)	**2020** Diet-induced obese mouse GI intolerance capped dose below glucagon engagement	**2021** Diabetic db/db mouse (chronic glucose lowering, Sanofi exendin-based dual agonists)	**2020** Healthy cynomolgus 4- and 8-week repeat-dose tox (cotadutide) Vomiting 17 % in patients
	type 2 diabetes mellitus SLC5A2 inhibitors (5)	**2012** ZDF rat renal glucose threshold	**2017** Single-species allometric scaling of rat pharmacokinetics to predict human clearance, volume…	**2023** db/db mouse and ZDF rat 4-week HbA1c	**2023** Rat 2-year carcinogenicity and chronic toxicology (SD rat, CD-1 mouse, dog)
	type 2 diabetes mellitus GCGR antagonists (4)	**2015** GCGR mAb in mice/monkeys	**2016** Glucagon-challenge hGCGR mice and rhesus monkeys; diabetic mice (small-molecule GRAs)	**2015** Gcgr-/- knockout mouse (genetic full antagonism)	**2022** GCGR mAb in mice/monkeys Transaminase, LDL-C, BP rises missed (T1D)
	type 2 diabetes mellitus GPR119 agonists (4)	**2014** Human primary colonic L-cells, GLP-1/PYY secretion ex vivo GLP-1/GIP/insulin flat; PYY rose	**2014** GPR119-agonist mice, OGTT	**2018** GPR119-agonist mice, OGTT Firuglipel failed HbA1c; comparator worked	**2017** GPR119-agonist mice, OGTT
	type 2 diabetes mellitus CCR2 antagonists (3)	**2013** Diet-induced obese / db/db mice dosed with a CCR2 antagonist	**2013** Human-CCR2 knock-in DIO / db/db mice dosed with CCX140-B	**2013** Diet-induced obese / db/db mice dosed with a CCR2 antagonist	**2005** Diet-induced obese / db/db mice dosed with a CCR2 antagonist
#2	**COPD** CHRM3 antagonists (6)	**2018** Human isolated bronchi: carbachol-evoked contraction, potency and washout offset	**2018** Human isolated bronchi: carbachol-evoked contraction, potency and washout offset	**2018** Human isolated bronchi: carbachol-evoked contraction, potency and washout offset	**2016** Rat bronchoprotection and antisialagogue; dog bronchospasm Stopped; preclinical unexplained
	COPD ADRB2 agonists (4)	**2016** Guinea-pig, human bronchi, dog bronchoprotection (abediterol)	**2016** Guinea-pig, human bronchi, dog bronchoprotection (abediterol)	**2016** Guinea-pig, human bronchi, dog bronchoprotection (abediterol) ADRB2 genotype not predictive	**2016** Guinea-pig, human bronchi, dog bronchoprotection (abediterol)
	COPD MAPK14 inhibitors (4)	**2018** Human alveolar macrophage and whole-blood LPS-TNF potency with rat inhaled lung PK (AZD7624)	**2020** Acumapimod PBPK DDI model, CYP3A4/P-gp inhibitors Lung target hit; 1 mg trial failed	**2019** Rodent LPS lung neutrophilia Acumapimod missed FEV1 primary twice	**2018** Human alveolar macrophage and whole-blood LPS-TNF potency with rat inhaled lung PK (AZD7624)
#3	**HIV** HIV RT (NNRTI) inhibitors (3)	**2012** HIV-1 culture vs wild-type and NNRTI-resistant virus, resistance selection, protein-adjusted IC90	**2020** In-silico doravirine PK model with rifabutin from the single-dose DDI, setting 100 mg twice daily	**2012** HIV-1 replication in cell culture (MT-4/PBMC) against wild-type and NNRTI-resistant virus	**2012** Rodent and monkey tox as filed for fosdevirine and doravirine Seizures no tox species predicted
	HIV HIV RT (NRTI) inhibitors (3)	**2013** NRTI-resistant HIV-1 in culture	**2021** Cell triphosphate persistence Threefold escalation still failed; abandoned	**2021** NRTI-resistant HIV-1 in culture Resistant-virus activity failed in patients	**2015** Chronic rat and monkey toxicology (6- and 9-month studies)
	HIV HIV integrase (INSTI) inhibitors (3)	**2011** HIV-1 replication in cell culture (MT-4/PBMC)	**2011** Protein-adjusted cell-culture potency as plasma trough target	**2012** In-vitro susceptibility of INSTI-resistant HIV-1 (site-directed mutant clones, fold-change)	**2025** Rat and rabbit embryo-fetal development studies Mouse NTD; human signal gone
	HIV HIV maturation (Gag) inhibitors (3)	**2021** Patient-derived and site-directed Gag-polymorphic HIV-1 in culture, resistance selection	**2023** Population PK/Emax model of Phase 2a monotherapy viral load; simulation picked Phase 2b doses	**2023** HIV-1 culture vs patient-derived and site-directed Gag-polymorphic viruses	**2025** Undisclosed repeat-dose tox Animal gastric lesions absent in patient biopsies
#4	**NSCLC** EGFR inhibitors (18)	**2017** T790M/sensitising-mutant NSCLC cells and mice	**2019** PC9 brain-metastasis mice; monkey osimertinib PET	**2023** Patient-derived lung cancer organoids EXCLAIM-2 fails primary PFS	**2015** hERG, rat and dog tox, dog telemetry QT prolongation the package cleared
	NSCLC ALK inhibitors (6)	**2019** Ba/F3 EML4-ALK L1196M cells and crizotinib-resistant patient-derived lines	**2018** EML4-ALK NSCLC xenografts (H2228/H3122, subcutaneous and intracranial)	**2019** Ba/F3 EML4-ALK L1196M cells and crizotinib-resistant patient-derived lines	**2018** Rat, monkey and dog toxicology
	NSCLC MET inhibitors (5)	**2021** METex14 and MET-amplified human tumour xenografts/PDX and HGF/MET allografts in mice	**2021** METex14 and MET-amplified human tumour xenografts/PDX and HGF/MET allografts in mice	**2023** Patient-derived lung cancer organoids, drug-sensitivity tested to steer the patient's next-line therapy	**2021** Tepotinib GLP rat and dog tox; pigmented-rat phototoxicity Oedema and ILD unpredicted
	NSCLC anti-IGF1R antibodies (4)	**2011** Tumour cells and xenografts given h7C10	**2005** A549 NSCLC xenografts in nude mice	**2011** A549 NSCLC xenografts in nude mice R1507 + erlotinib: no PFS gain	**2021** Cynomolgus repeat-dose tox (R1507) Monkeys missed hyperglycemia
	NSCLC anti-PDCD1 antibodies (4)	**2023** Toripalimab in human-PBMC NSG mice, T-cell expansion and effector-memory as PD readout	**2023** Cynomolgus pharmacology and repeat-dose tox, PD-1 receptor occupancy (toripalimab, dostarlimab)	**2023** MC38 tumours in NSG mice engrafted with allogeneic human CD8+ T cells, dosed with toripalimab	**2021** PD-1 knockout mice Monkeys silent; patients had irAEs
	NSCLC pan-HER inhibitors (4)	**2011** EGFR del19/L858R NSCLC lines and HCC827 xenografts	**2017** EGFR del19/L858R NSCLC lines and HCC827 xenografts 16 mg margin: 72% dose intensity	**2017** EGFR del19/L858R NSCLC lines and HCC827 xenografts 0/11 exon-20 responses; HER2…	**2020** Exon 20/ERBB2-mutant patient lines, xenografts (tarloxotinib) PDX silent on grade 3 diarrhoea
	NSCLC HSP90AA1/HSP90AB1 inhibitors (3)	**2013** NSCLC lines and xenografts	**2013** NSCLC lines and xenografts Exposure reached; the class still failed	**2018** EGFR exon 20 insertion cell models (Ba/F3 and NSCLC lines) Docetaxel phase 3 failed, HR 1.11	**2013** Mouse xenografts, cardiac and liver readouts (ganetespib)
	NSCLC VEGFR multikinase inhibitors (3)	**2022** Mouse Lewis lung carcinoma: famitinib or lenvatinib added to radiotherapy and anti-PD-1/PD-L1	**2022** Mouse Lewis lung carcinoma: famitinib or lenvatinib added to radiotherapy and anti-PD-1/PD-L1	**2019** RET-fusion lung, thyroid and Ba/F3 cells; NIH3T3 xenografts Three phase 3s failed; HR 1.10	**2019** GLP repeat-dose toxicology in rat plus monkey (and dog)
	NSCLC anti-CTLA4 antibodies (3)	**2010** B16, MC38 and CT26 mouse tumours, anti-CTLA-4 plus anti-PD-1	**2018** B16, MC38 and CT26 mouse tumours, anti-CTLA-4 plus anti-PD-1 10 mg/kg reached; failed; dropped	**2020** B16, MC38 and CT26 mouse tumours, anti-CTLA-4 plus anti-PD-1 Nivo-ipi-chemo OS met, HR 0.69	**2011** CTLA-4 knockout mice Monkeys missed fatal immune tox
#5	**migraine disorder** CGRP receptor antagonists (4)	**2014** Rhesus capsaicin-induced dermal vasodilation (CIDV)	**2014** Rhesus CIDV PK/PD target Predicted 20 mg; needed 200 mg	**2013** Primate CGRP-induced facial blood flow (rimegepant / BMS-927711)	**2026** Mouse MCAO stroke, gepant pretreatment Animal tox missed MK-3207 liver
	migraine disorder anti-CGRP ligand antibodies (3)	**2015** Cynomolgus capsaicin-induced dermal blood flow, PK/PD-translated to humans	**2015** Cynomolgus capsaicin-induced dermal blood flow, PK/PD-translated to humans	**2014** Rat evoked neurogenic vasodilation (skin flare, middle meningeal artery)	**2022** Rat blood pressure, heart rate on anti-CGRP mAb; monkey safety Postmarketing hypertension warning
#6	**Alzheimer disease** BACE1 inhibitors (4)	**2015** PDAPP mice and cannulated beagle dogs (brain/CSF Aβ)	**2019** Dog CSF Aβ PK/PD scaled to humans Target reached, no benefit	**2019** A673T variant carriers Genetics said protect; drug worsened	**2019** Mice chronically dosed with BACE1 inhibitors Liver injury missed by tox
	Alzheimer disease anti-MAPT antibodies (4)	**2023** Tau mouse interstitial fluid Tau captured; CSF Abeta42 flat	**2023** Tau mouse interstitial fluid ISF target met in CSF; CDR-SB unchanged	**2022** Human AD-brain PHF-tau seeding in HEK293 biosensor cells AD-seed removal did not translate	**2021** Cynomolgus monkeys and tau-transgenic mice (PK/PD; toxicology species unpublished)